

Table 3: Learning objectives (Theory) of Course

| Paper 1 Fundamental Dravyaguna                                        |                                                                                             |                  |                                                       |                                              |                  |                                        |                           |            |                   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------|----------------------------------------------|------------------|----------------------------------------|---------------------------|------------|-------------------|
| A3<br>Course<br>outcome                                               | B3<br>Learning Objective (At the end of the session, the students should be able to)        | C3<br>Domain/sub | D3<br>Must to know / desirable to know / Nice to know | E3<br>Level Does/ Shows how/ Knows how/ Know | F3<br>T-L method | G3<br>Assessment (Refer abbreviations) | H3<br>Formative/summative | I3<br>Term | J3<br>Integration |
| Topic 1 1.Dravyaguna Vigyana (Lecture :1 hours, Non lecture: 1 hours) |                                                                                             |                  |                                                       |                                              |                  |                                        |                           |            |                   |
| CO1,CO3                                                               | Define Dravyaguna Vigyana.                                                                  | CK               | MK                                                    | K                                            | L&PP<br>T        | P-VIVA                                 | F&S                       | I          |                   |
| CO1,CO3                                                               | Explain the role of Dravyaguna Vigyana in clinical practice.                                | CC               | MK                                                    | KH                                           | L&G<br>D         | P-VIVA,C-INT                           | F&S                       | I          |                   |
| Topic 2 2.Dravya (Lecture :5 hours, Non lecture: 4 hours)             |                                                                                             |                  |                                                       |                                              |                  |                                        |                           |            |                   |
| CO1,CO5,CO8                                                           | Classify Dravya based on Utpatti, Yonibheda, Prayogabheda & Prabhavbheda.                   | CK               | MK                                                    | K                                            | L&PP<br>T,LS     | T-OBT,P-VIVA,P-POS                     | F&S                       | I          |                   |
| CO1,CO5,CO8                                                           | Classify Dravya based on Doshagnabheda, Rasabheda, Karmbheda (Dashemani & Sushrutokta Gana) | CK               | MK                                                    | K                                            | L&PP<br>T,LS     | T-OBT,P-VIVA,P-POS                     | F&S                       | I          |                   |
| CO1,CO3                                                               | Describe Panchabhoutikatwa of Dravya.                                                       | CC               | MK                                                    | KH                                           | DIS,P<br>ER      | T- EW,T-OBT,P-VIVA                     | F&S                       | I          |                   |
| Topic 3 3. Guna (Lecture :4 hours, Non lecture: 2 hours)              |                                                                                             |                  |                                                       |                                              |                  |                                        |                           |            |                   |
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| CO1,CO2,CO3                                                     | Define Guna and describe its Panchabhoutikatva.                                             | CK  | MK | K  | L&PP<br>T,DIS<br>,LS      | T- EW,P-VIV<br>A,PRN,QZ         | F&S | I |  |
| CO1                                                             | State characteristics of Gurvadi Guna.                                                      | CK  | MK | K  | L&PP<br>T,SD<br>L         | P-VIVA,PRN,<br>QZ               | F&S | I |  |
| CO1,CO2,CO3                                                     | Discuss Gurvadiguna in context to its Karma on Dosha, Dhatu and Mala with examples.         | CC  | MK | KH | L&PP<br>T,L&<br>GD,F<br>C | T- EW,P-VIV<br>A,PRN,OSPE       | F&S | I |  |
| CO1,CO2,CO3                                                     | Enumerate & Summarize, clinical application of Gurvadi Guna with research updates.          | CC  | MK | KH | L&G<br>D,BL,<br>LS        | T- EW,P-<br>VIVA,PRN            | F&S | I |  |
| CO1,CO2,CO3                                                     | Enlist Paradiguna with examples.                                                            | CK  | MK | K  | L&G<br>D,PE<br>R          | T-EMI,P-VIV<br>A,PRN,P-POS      | F&S | I |  |
| CO1,CO2,CO3                                                     | Interpret clinical applications of Paradiguna.                                              | CAP | MK | KH | L&PP<br>T,DIS             | T-EMI,P-VIV<br>A,PRN,QZ         | F&S | I |  |
| CO1,CO2,CO3                                                     | Discuss research updates of Paradiguna.                                                     | CC  | MK | KH | L,TP<br>W,LS              | T-EMI,P-VIV<br>A,PRN,CR-<br>RED | F&S | I |  |
| <b>Topic 4 4. Rasa</b> (Lecture :7 hours, Non lecture: 4 hours) |                                                                                             |     |    |    |                           |                                 |     |   |  |
| CO1,CO2,CO3                                                     | Define Rasa and interpret Meaning of “Rasa” in various contexts.                            | CK  | MK | K  | L&G<br>D                  | P-VIVA,PUZ,<br>M-CHT,INT        | F&S | I |  |
| CO1,CO2                                                         | Classify and compare shadarasa in relative correlation with taste of chemical constituents. | CK  | MK | K  | L&PP<br>T,ED              | P-VIVA,PUZ,<br>M-CHT,INT        | F&S | I |  |



|                                                                   |                                                                                      |     |    |    |                     |                                      |     |   |  |
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| CO1,CO2                                                           | Define Anurasa.                                                                      | CK  | MK | K  | L,LS                | T-EMI,P-VIVA,INT                     | F&S | I |  |
| CO1,CO2                                                           | Interpret relevance of Anurasa in clinical practice.                                 | CAP | MK | KH | L&G D,FC            | T-EMI,P-VIVA,PUZ,INT                 | F&S | I |  |
| CO1,CO2                                                           | Describe with justification Rasa sevanakrama as Aushadha.                            | CC  | MK | KH | L&PP T,DIS          | T-EMI,T- EW, P-VIVA,PRN, M-CHT,M-POS | F&S | I |  |
| <b>Topic 5 5. Vipaka</b> (Lecture :3 hours, Non lecture: 1 hours) |                                                                                      |     |    |    |                     |                                      |     |   |  |
| CO1                                                               | Define Vipaka.                                                                       | CK  | MK | K  | L                   | P-VIVA                               | F&S | I |  |
| CO1                                                               | State Vipaka lakshana.                                                               | CK  | MK | K  | L&G D               | P-VIVA,INT                           | F&S | I |  |
| CO1                                                               | Describe the action of Trividha Vipaka on Dosha, Dhatu and Mala with examples.       | CC  | MK | KH | L&PP T,TP W,FC      | T- EW, P-VIVA,PRN                    | F&S | I |  |
| CO1,CO3                                                           | Interpret clinical application of Vipaka.                                            | CAP | MK | KH | L&PP T,DIS ,SDL, LS | P-VIVA,QZ                            | F&S | I |  |
| CO2                                                               | State Research updates of Vipaka.                                                    | CK  | MK | K  | L&PP T              | P-VIVA,CR-RED                        | F&S | I |  |
| CO1                                                               | Explain Vipakopalabdh (Determination of Vipaka) and Taratamya (Degree of variation). | CC  | MK | KH | L&PP T,FC           | P-VIVA,PUZ, M-CHT                    | F&S | I |  |
| <b>Topic 6 6. Virya</b> (Lecture :2 hours, Non lecture: 2 hours)  |                                                                                      |     |    |    |                     |                                      |     |   |  |
| CO1                                                               | Define the lakshanas of Virya.                                                       | CK  | MK | K  | L,SD                | P-VIVA,INT                           | F&S | I |  |



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| CO1,CO2                                                                                                     | Discuss research updates of Prabhava.                                                                            | CC  | MK | KH | L&G<br>D,SD<br>L          | T- EW,P-VIV<br>A,CR-RED     | F&S | I |  |
| <b>Topic 8 8. Interrelation of Rasa-Guna-Virya-Vipaka-Prabhava</b> (Lecture :1 hours, Non lecture: 2 hours) |                                                                                                                  |     |    |    |                           |                             |     |   |  |
| CO1                                                                                                         | Interpret the interrelation of Rasa-Guna-Virya-Vipaka-Prabhava with respect to their strength- Pharmacodynamics. | CAP | MK | KH | L&PP<br>T,TP<br>W,PL      | P-VIVA,CL-<br>PR            | F&S | I |  |
| <b>Topic 9 9. Karma</b> (Lecture :9 hours, Non lecture: 5 hours)                                            |                                                                                                                  |     |    |    |                           |                             |     |   |  |
| CO1                                                                                                         | Define Karma.                                                                                                    | CK  | MK | K  | L                         | T-OBT,P-<br>VIVA,QZ         | F&S | I |  |
| CO1                                                                                                         | Discuss Karma lakshana.                                                                                          | CC  | MK | KH | L&G<br>D                  | T-OBT,P-<br>VIVA,QZ         | F&S | I |  |
| CO1,CO2,CO<br>4,CO5                                                                                         | Explain Deepan karma in relation with Appetizers.                                                                | CC  | MK | KH | L&G<br>D,BL               | T- EW,T-OBT<br>,P-VIVA,QZ   | F&S | I |  |
| CO1,CO2,CO<br>3,CO4                                                                                         | Critically analyze Deepan dravyas with its gunas and application in various clinical conditions.                 | CAP | MK | KH | L&PP<br>T,PB<br>L,GB<br>L | T-OBT,P-<br>VIVA,QZ<br>,DEB | F&S | I |  |
| CO1,CO2,CO<br>3,CO4                                                                                         | Explain Pachan karma in relation with Digestives.                                                                | CC  | MK | KH | L&PP<br>T                 | T-OBT,P-<br>VIVA,QZ         | F&S | I |  |
| CO1,CO2,CO<br>3,CO4                                                                                         | Critically analyze Pachan dravyas with gunas and discuss its application in various clinical conditions.         | CAN | MK | KH | L&PP<br>T,PB<br>L,PE<br>R | T-EMI,T-<br>OBT,P-VIVA      | F&S | I |  |
| CO1,CO2,CO                                                                                                  | Explain Samshodhan karma in relation with contemporary                                                           | CC  | MK | KH | L<br>VC                   | T-OBT,P-                    | F&S | I |  |



| 4               | pharmacological action.                                                        |    |    |    |  | ,PL            | VIVA,QZ                 |     |   |  |
|-----------------|--------------------------------------------------------------------------------|----|----|----|--|----------------|-------------------------|-----|---|--|
| CO1,CO3,CO4     | Discuss Samshohan karma and its clinical application.                          | CC | MK | KH |  | PBL,RP,PER     | T-EW,T-CS,T-OBT,P-VIVA  | F&S | I |  |
| CO1,CO2,CO3,CO4 | Explain Samshamana karma in relation with contemporary pharmacological action. | CC | MK | KH |  | L&GD,BL        | T-EMI,T-OBT,P-VIVA      | F&S | I |  |
| CO1,CO2,CO3,CO4 | Describe Samshamana karma and its clinical application.                        | CC | MK | KH |  | L&PP T,CB L,PL | T-EMI,T-CS,P-VIVA,QZ    | F&S | I |  |
| CO1,CO2,CO3     | Explain Anuloman karma in relation with Carminative.                           | CC | MK | KH |  | L_VC ,DIS      | T-EMI,T-OBT ,P-VIVA,DEB | F&S | I |  |
| CO1,CO3         | Describe Anuloman karma and its clinical application.                          | CC | MK | KH |  | L&PP T,PB L,LS | T-OBT,P-VIVA,QZ         | F&S | I |  |
| CO1,CO2,CO3,CO4 | Explain Sransana karma in relation with contemporary pharmacological action.   | CC | MK | KH |  | L&PP T         | T-EMI,T-OBT ,P-VIVA,DEB | F&S | I |  |
| CO1,CO3         | Discuss Sransan karma and its clinical application.                            | CC | MK | KH |  | L&PP T,PB L,LS | T-OBT,P-VIVA,QZ ,DEB    | F&S | I |  |
| CO1,CO2,CO3,CO4 | Explain Bhedana karma in relation with strong laxative action                  | CC | MK | KH |  | L_VC ,FC       | T-EMI,T-OBT ,P-VIVA,DEB | F&S | I |  |
| CO1,CO3         | Discuss Bhedan karma and its clinical application.                             | CC | MK | KH |  | L&PP T,DIS     | T-OBT,P-VIVA,QZ         | F&S | I |  |
| CO1,CO2,CO3,CO4 | Explain Rechana karma in relation with Purgative.                              | CC | MK | KH |  | L&PP T         | T-EMI,T-OBT,P-VIVA      | F&S | I |  |

|                     |                                                                                         |    |    |    |                      |                             |     |   |  |
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| CO1,CO3             | Discuss Rechana karma and its clinical application.                                     | CC | MK | KH | L&PP<br>T,DIS        | T-OBT,P-<br>VIVA,QZ         | F&S | I |  |
| CO1,CO3             | Discuss Chhedana karma and its clinical application.                                    | CC | MK | KH | L&PP<br>T,CB<br>L,LS | T-EMI,T-<br>OBT,P-VIVA      | F&S | I |  |
| CO1,CO2,CO<br>3,CO4 | Explain Lekhana karma in relation with contemporary pharmacological action.             | CC | MK | KH | L&G<br>D,BL          | T-EMI,T-<br>OBT,P-VIVA      | F&S | I |  |
| CO1,CO3             | Discuss Lekhana karma and its clinical application.                                     | CC | MK | KH | L&PP<br>T,DIS<br>,PL | T-OBT,P-<br>VIVA,QZ<br>,DEB | F&S | I |  |
| CO1,CO2,CO<br>3,CO4 | Explain Grahi karma in relation with contemporary pharmacological action bowel binding. | CC | MK | KH | L VC                 | T-EMI,T-<br>OBT,P-VIVA      | F&S | I |  |
| CO1,CO3             | Discuss of Grahi karma and its clinical application.                                    | CC | MK | KH | L&G<br>D,TB<br>L     | T-OBT,P-<br>VIVA,QZ         | F&S | I |  |
| CO1,CO2,CO<br>3,CO4 | Explain Stambhana karma in relation with contemporary pharmacological action.           | CC | MK | KH | L&PP<br>T,FC         | T-EMI,T-OBT<br>,P-VIVA,DEB  | F&S | I |  |
| CO1,CO3             | Discuss Stambhana karma and its clinical application.                                   | CC | MK | KH | L&G<br>D,PB<br>L     | T-EMI,T-OBT<br>,P-VIVA,DEB  | F&S | I |  |
| CO1                 | Discuss Madakari karma with examples.                                                   | CC | MK | KH | L,DIS                | T-OBT,P-<br>VIVA,QZ         | F&S | I |  |
| CO1,CO3             | Discuss Pramathi karma and its clinical application.                                    | CC | MK | KH | L&PP<br>T,DIS        | T-EMI,T-<br>OBT,P-VIVA      | F&S | I |  |
| CO1                 | Discuss Abhishyandi karma with examples.                                                | CK | MK | K  | L,DIS                | T-EMI,T-OBT                 | F&S | I |  |







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|     | Bronchodilators, Aerosols/ Inhalants, Expectorants and Antitussives Drugs                                                                                                                                                                                           |    |    |  |  | L,BL |                             |                            |     |     |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting on Cardiovascular Drugs as Antihypertensive and Antianginal Drugs                                                                                                           | CC | MK |  |  | KH   | L_VC<br>_FC                 | T-OBT,P-<br>VIVA,PRN       | F&S | III |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting on Kidney as Diuretics                                                                                                                                                      | CC | MK |  |  | KH   | L&PP<br>T,TP<br>W,BL        | T-OBT,P-<br>VIVA,QZ        | F&S | III |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting as Haematinics, Coagulants, Anticoagulants and Hypolipidaemic Drugs                                                                                                         | CC | MK |  |  | KH   | L&PP<br>T,PB<br>L,BL        | T- EW,T-<br>OBT,P-VIVA     | F&S | III |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting on Gastrointestinal tract as Antacid, Carminatives, Digestants, Antiemetics, Laxatives, Antidiarrhoeal and Hepatoprotective Drugs                                           | CC | MK |  |  | KH   | L&PP<br>T,L&<br>GD,L<br>_VC | T- EW,T-OBT<br>,P-VIVA,QZ  | F&S | III |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting as Antibiotics and Antitubercular.                                                                                                                                          | CC | MK |  |  | KH   | L&PP<br>T,FC,<br>GBL        | T- EW,P-<br>VIVA,QZ        | F&S | III |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting as <b>Antifungal, Antiviral, Antimalarial and Anthelmintic Drugs</b>                                                                                                        | CC | MK |  |  | KH   | L&PP<br>T,PB<br>L,GB<br>L   | T- EW,T-OBT<br>,P-VIVA,QZ  | F&S | III |  |
| CO9 | Define, Describe mode of action & Discuss types with examples of following drugs acting on Hormones and Related Drugs as Thyroid Hormone, Thyroid Inhibitors, Insulins, Oral Antidiabetic, Hormonal Contraceptives, Uterine Stimulants and Uterine Relaxants Drugs. | CC | MK |  |  | KH   | L&PP<br>T,BL,<br>PER        | T- EW,T-OBT<br>,P-VIVA,PRN | F&S | III |  |

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| CO9                                                                        | Define, Describe mode of action & Discuss types with examples of following drugs acting on as Antiseptics , Disinfectants, Vaccines, Vitamins, Water imbalance and IV | CC | MK | KH | L&PP<br>T,BL,<br>PER  | T- EW,T-<br>OBT,P-VIVA              | F&S | III |  |
| <b>Topic 12 12. Mishraka Gana</b> (Lecture :6 hours, Non lecture: 2 hours) |                                                                                                                                                                       |    |    |    |                       |                                     |     |     |  |
| CO1,CO3                                                                    | Explain Brihatpanchamoola composition.                                                                                                                                | CC | MK | KH | L&PP<br>T,ED<br>U     | T- EW,P-VIV<br>A,P-POS,QZ<br>,M-CHT | F&S | III |  |
| CO1,CO3                                                                    | Describe Guna karma and combined therapeutic effect of Brihatpanchamoola.                                                                                             | CC | MK | KH | L&PP<br>T,DIS<br>,TBL | T- EW,P-VIV<br>A,P-POS,QZ<br>,M-CHT | F&S | III |  |
| CO1,CO3                                                                    | Explain Laghupanchamoola composition.                                                                                                                                 | CC | MK | KH | L&G<br>D              | T- EW,P-VIV<br>A,P-POS,QZ<br>,M-CHT | F&S | III |  |
| CO1,CO3                                                                    | Describe Guna karma and combined therapeutic effect of Laghupanchamoola.                                                                                              | CC | MK | KH | L&PP<br>T,DIS<br>,PL  | T- EW,P-VIV<br>A,P-POS,QZ           | F&S | III |  |
| CO1,CO3                                                                    | Explain Vallipanchamoola composition.                                                                                                                                 | CC | MK | KH | L&PP<br>T             | T- EW,P-VIV<br>A,P-POS,QZ<br>,M-CHT | F&S | III |  |
| CO1,CO3                                                                    | Describe Guna karma and combined therapeutic effect of Vallipanchamoola.                                                                                              | CC | MK | KH | L&G<br>D,PE<br>R      | T- EW,P-<br>VIVA,QZ                 | F&S | III |  |
| CO1,CO3                                                                    | Explain Kantakapanchamoola composition.                                                                                                                               | CC | MK | KH | L&PP<br>T,BS,         | T- EW,P-<br>VIVA,QZ                 | F&S | III |  |

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| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Kantakapanchamoola. | CC | MK |  | KH | TPW            | P-POS,QZ ,M-CHT         | F&S | III |  |  |
| CO1,CO3 | Explain Trinapanchamoola composition.                                      | CC | MK |  | KH | L_VC,PER       | P-VIVA,P-EXAM,QZ        | F&S | III |  |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Trinapanchamoola.   | CC | MK |  | KH | L&PP T,CB L,PL | T- EW,P-VIVA,M-CHT      | F&S | III |  |  |
| CO1,CO3 | Explain Panchavalkala composition.                                         | CC | MK |  | KH | L&PP T,BL,GBL  | P-VIVA,P-POS,QZ         | F&S | III |  |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Panchavalkala.      | CC | MK |  | KH | L&PP T,FC      | T-CS,T-OBT, P-VIVA,QZ   | F&S | III |  |  |
| CO1,CO3 | Explain Triphala composition                                               | CC | MK |  | KH | L,DIS          | T- EW,P-VIVA,P-POS      | F&S | III |  |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Triphala.           | CC | MK |  | KH | L&PP T,PE R    | T-OBT,P-VIV A,P-EXAM,QZ | F&S | III |  |  |
| CO1,CO3 | Explain Trikatu composition.                                               | CC | MK |  | KH | L&PP T,PL      | T-EMI,P-VIVA,QZ         | F&S | III |  |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Trikatu             | CC | MK |  | KH | L&PP T,PL      | T-OBT,P-VIVA,P-POS      | F&S | III |  |  |
| CO1,CO3 | Explain Trimada composition                                                | CC | MK |  | KH | L,DIS          | T- EW,P-VIVA,QZ         | F&S | III |  |  |

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| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Trimada.     | CC | MK | KH | L&PP<br>T,DIS<br>,TUT    | T- EW,P-<br>VIVA,QZ               | F&S | III |  |
| CO1,CO3 | Explain Chaturusana composition.                                    | CC | MK | KH | L&PP<br>T                | T-OBT,P-<br>VIVA,P-POS            | F&S | III |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Chaturusana. | CC | MK | KH | L&PP<br>T,FC             | P-VIVA,P-<br>POS,QZ               | F&S | III |  |
| CO1,CO3 | Explain Panchakola composition.                                     | CC | MK | KH | L&PP<br>T,L_<br>VC       | P-VIVA,QZ                         | F&S | III |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Panchakola.  | CC | MK | KH | L&PP<br>T,DIS            | T- EW,T-CS,P-<br>VIVA,QZ          | F&S | III |  |
| CO1,CO3 | Explain Shadusana composition.                                      | CC | MK | KH | L&PP<br>T,PL             | T-OBT,P-<br>VIVA,P-POS            | F&S | III |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Shadusana    | CC | MK | KH | L&PP<br>T,FC             | T- EW,P-<br>VIVA,QZ               | F&S | III |  |
| CO1,CO3 | Explain Chaturbeeja composition.                                    | CC | MK | KH | L&PP<br>T,L_<br>VC       | T-OBT,P-VIV<br>A,P-EXAM           | F&S | III |  |
| CO1,CO3 | Describe Guna karma and combined therapeutic effect of Chaturbeeja  | CC | MK | KH | L&PP<br>T,PE<br>R        | T- EW,P-<br>VIVA,QZ               | F&S | III |  |
| CO1,CO3 | Explain Trijataka composition.                                      | CC | MK | KH | L_<br>VC<br>,DIS,<br>SDL | T-OBT,P-VIV<br>A,P-EXAM,P-<br>POS | F&S | III |  |



|                                                                                                                                |                                                                       |    |    |    |                      |                           |     |     |  |
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| CO1,CO3                                                                                                                        | Describe Guna karma and combined therapeutic effect of Trijataka      | CC | MK | KH | L&PP<br>T,PL         | T- EW,P-<br>VIVA,P-POS    | F&S | III |  |
| CO1,CO3                                                                                                                        | Explain Chaturajataka composition.                                    | CC | MK | KH | L_VC<br>,PER         | T-OBT,P-<br>VIVA,QZ       | F&S | III |  |
| CO1,CO3                                                                                                                        | Describe Guna karma and combined therapeutic effect of Chaturajataka. | CC | MK | KH | L&PP<br>T,DIS        | T- EW,P-<br>VIVA,P-POS    | F&S | III |  |
| CO1,CO3                                                                                                                        | Explain Panchatikta composition.                                      | CC | MK | KH | L&PP<br>T,FC         | T-OBT,P-VIV<br>A,P-REC,QZ | F&S | III |  |
| CO1,CO3                                                                                                                        | Describe Guna karma and combined therapeutic effect of Panchatikta.   | CC | MK | KH | L&PP<br>T,PL         | T- EW,P-<br>VIVA,P-POS    | F&S | III |  |
| CO1,CO3                                                                                                                        | Explain Chaturbhadra composition.                                     | CK | MK | K  | L,DIS                | T-OBT,P-<br>VIVA,QZ       | F&S | III |  |
| CO1,CO3                                                                                                                        | Describe Guna karma and combined therapeutic effect of Chaturbhadra.  | CC | MK | KH | L&PP<br>T,FC         | T-EMI,P-<br>VIVA,QZ       | F&S | III |  |
| CO1,CO3                                                                                                                        | Explain Trikarshika composition.                                      | CC | MK | KH | L,DIS                | T-OBT,P-VIV<br>A,P-EXAM   | F&S | III |  |
| CO1,CO3                                                                                                                        | Describe Guna karma and combined therapeutic effect of Trikarshika.   | CC | MK | KH | L&PP<br>T,DIS<br>,PL | T-OBT,P-<br>VIVA,QZ       | F&S | III |  |
| <b>Topic 13 13. Nomenclature of dravya as per Nighantu, Vedic taxonomy and Botany</b> (Lecture :0 hours, Non lecture: 2 hours) |                                                                       |    |    |    |                      |                           |     |     |  |
| CO1                                                                                                                            | Describe the Nomenclature of dravya as per Raj Nighantu.              | CC | MK | KH | L&PP<br>T,FC,<br>RP  | P-VIVA,PRN,<br>P-ID,QZ    | F&S | III |  |
| CO1                                                                                                                            | Explain the Nomenclature of dravya as per Vedic taxonomy.             | CC | MK | KH | L&PP<br>T            | P-VIVA,P-<br>ID,QZ        | F&S | III |  |

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| CO1                                                                                                                                                                                                                    | Discuss the Nomenclature of dravya as per botany.                                                                                                      | CC | MK | KH | L&PP<br>T,BL,<br>PER      | P-VIVA,P-<br>ID,QZ  | F&S | III |  |
| <b>Topic 14 14. Prashasta Bhesaja, Bhesaja Pariksha and drug evaluation method with correlation as per Pharmacognosy</b> (Lecture :1 hours, Non lecture: 2 hours)                                                      |                                                                                                                                                        |    |    |    |                           |                     |     |     |  |
| CO1,CO5                                                                                                                                                                                                                | Describe Prashasta Bhesaja.                                                                                                                            | CC | MK | KH | L&G<br>D,PL               | P-VIVA,QZ           | F&S | III |  |
| CO1,CO5                                                                                                                                                                                                                | Explain Bhesaja Pariksha of Charaka samhita vimana sthana.                                                                                             | CC | MK | KH | L&G<br>D,SD<br>L,LS       | P-VIVA,QZ<br>,M-CHT | F&S | III |  |
| CO1,CO5                                                                                                                                                                                                                | Discuss drug evaluation method in correlation with Pharmacognosy.                                                                                      | CC | MK | KH | L&PP<br>T,DIS             | P-VIVA,QZ           | F&S | III |  |
| <b>Topic 15 15. Dravyasangrahana and Drug collection methods as per GFCP (Good Field collection practices)</b> (Lecture :1 hours, Non lecture: 0 hours)                                                                |                                                                                                                                                        |    |    |    |                           |                     |     |     |  |
| CO1,CO2                                                                                                                                                                                                                | Discuss Dravyasangrahana and Drug collection methods as per GFCP (Good Field collection practices).                                                    | CC | DK | KH | L&PP<br>T,PE<br>R         | P-VIVA,PUZ,<br>CHK  | F&S | III |  |
| CO1,CO2                                                                                                                                                                                                                | Discuss Drug collection methods as per GFCP (Good Field collection practices).                                                                         | CC | DK | KH | L&PP<br>T,DIS<br>,PrBL    | P-VIVA,PUZ,<br>CHK  | F&S | III |  |
| <b>Topic 16 16. GCP (Good cultivation practices), seed bank, conservation of medicinal plants, knowledge about RET (Rear, Endangered &amp; Threatened ) medicinal plants.</b> (Lecture :1 hours, Non lecture: 0 hours) |                                                                                                                                                        |    |    |    |                           |                     |     |     |  |
| CO2                                                                                                                                                                                                                    | Discuss good cultivation practices, seed bank, conservation of medicinal plants, knowledge about RET (Rear, Endangered & Threatened) medicinal plants. | CC | DK | KH | L&PP<br>T,L_<br>VC,F<br>C | P-VIVA,M-<br>POS    | F&S | III |  |



|                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |
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| <b>Topic 17 17. Abhava Pratinidhi Dravya (substitutes) (Lecture :1 hours, Non lecture: 1 hours)</b>                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |
| CO1,CO7                                                                                                                                                                                                                                                                                                                  | Discuss the concept of Abhava pratinidhi dravya (substitutes) as per Bhavaprakasha.                                                                                                                                                                                  | CC | MK | KH | L&G<br>D,BS,<br>EDU       | P-VIVA,QZ                | F&S<br>III         |
| <b>Topic 18 18. Classifications and techniques of aqueous and alcoholic extracts (Lecture :0 hours, Non lecture: 2 hours)</b>                                                                                                                                                                                            |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |
| CO2,CO6                                                                                                                                                                                                                                                                                                                  | Appraise the techniques of aqueous and alcoholic extracts of medicinal plants.                                                                                                                                                                                       | CC | DK | KH | L_V<br>_D                 | P-<br>VIVA,DOPS          | F&S<br>III         |
| <b>Topic 19 19. Adverse drug reaction and Pharmacovigilance with recent updates (Lecture :1 hours, Non lecture: 2 hours)</b>                                                                                                                                                                                             |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |
| CO2,CO8                                                                                                                                                                                                                                                                                                                  | Explain adverse drug reaction and pharmacovigilance in ayurveda with recent updates.                                                                                                                                                                                 | CC | MK | KH | L&PP<br>T,L&<br>GD,L<br>S | P-VIVA,PRN,<br>QZ ,M-CHT | F&S<br>III<br>V-RS |
| <b>Topic 20 20. NMPB (National Medicinal Plant Board), CCRAS (Central Council of Research in Ayurveda Sciences), API ( Ayurvedic Pharmacopeia of India), GCTM ( Global Centre for Traditional Medicine), PCIMH ( Pharmacopeia Commission of Indian Medicine and Homeopathy) (Lecture :1 hours, Non lecture: 0 hours)</b> |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |
| CO2                                                                                                                                                                                                                                                                                                                      | State NMPB (National Medicinal Plant Board), CCRAS (Central Council of Research in Ayurveda Sciences), API ( Ayurvedic Pharmacopeia of India), GCTM ( Global Centre for Traditional Medicine) and PCIMH (Pharmacopeia Commission of Indian Medicine and Homeopathy). | CK | NK | K  | SDL                       | P-VIVA,QZ<br>,INT        | F&S<br>III         |
| <b>Topic 21 21. Vrikshayurveda and Ethno-medicine (Lecture :1 hours, Non lecture: 1 hours)</b>                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |
| CO2                                                                                                                                                                                                                                                                                                                      | Explain concept of Vrikshayurveda and Ethno medicine.                                                                                                                                                                                                                | CC | NK | KH | L&G<br>D,ML<br>,PER       | P-VIVA,CR-<br>RED        | F&S<br>III         |
| <b>Topic 22 22. Network pharmacology and Bioinformatics (Lecture :1 hours, Non lecture: 1 hours)</b>                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                      |    |    |    |                           |                          |                    |

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|-----|---------------------------------------------------|----|----|----|---------------------|-------------------|-----|-----|--|
| CO2 | Describe Network Pharmacology and Bioinformatics. | CC | NK | KH | L&G<br>D,SD<br>L,LS | P-VIVA,CR-<br>RED | F&S | III |  |
|-----|---------------------------------------------------|----|----|----|---------------------|-------------------|-----|-----|--|

| Paper 2 Applied Dravyaguna                                                                                                              |                                                                                                                                             |                   |                                                       |                                              |                                    |                                        |                            |            |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------|----------------------------------------------|------------------------------------|----------------------------------------|----------------------------|------------|-------------------|
| A3<br>Course outcome                                                                                                                    | B3<br>Learning Objective (At the end of the session, the students should be able to)                                                        | C3<br>Doma in/sub | D3<br>Must to know / desirable to know / Nice to know | E3<br>Level Does/ Shows how/ Knows how/ Know | F3<br>T-L method                   | G3<br>Assessment (Refer abbreviations) | H3<br>Formative/ summative | I3<br>Term | J3<br>Integration |
| Topic 1 1. Bheshajavacharaniya (Criteria’s to be considered for selection of drugs in vyadhis) (Lecture :1 hours, Non lecture: 6 hours) |                                                                                                                                             |                   |                                                       |                                              |                                    |                                        |                            |            |                   |
| CO5                                                                                                                                     | Interpret the selection of appropriate drugs in different vyadhis as per criteria’s mentioned in Bheshajavacharaniya (as per As. Sa. Su 23) | CAP               | MK                                                    | KH                                           | L&PP<br>T,CB<br>L,PrB<br>L,SD<br>L | T- EW,P-VIVA,QZ                        | F&S                        | II         | H-RN ,H-SW        |
| Topic 2 2.1 Dravya (Drug) Nama-Guna-Karma Jnana (Lecture :45 hours, Non lecture: 10 hours)                                              |                                                                                                                                             |                   |                                                       |                                              |                                    |                                        |                            |            |                   |
| CO3                                                                                                                                     | Specify useful parts with its Rasapanchaka of following drugs.                                                                              | CK                | MK                                                    | K                                            | L&G<br>D,CB<br>L,FC                | T-OBT,P-VIVA                           | F&S                        | II         |                   |
| CO4                                                                                                                                     | Describe karma,agryakarma and dosha karma of following drugs.                                                                               | CC                | MK                                                    | KH                                           | L&PP                               | T-EMI,T-OBT                            | F&S                        | II         |                   |



|     | various Srotas and vyadhiavastha (Stage).                                                                                                                                             |     |    |    |  | T,DIS<br>,CBL         | VIVA,QZ                    |     |     |               |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|--|-----------------------|----------------------------|-----|-----|---------------|
| CO5 | Indicate the Kalpana(dosage form), Matra (Dose), Anupana (Vehicle), Marga (Route), Sevana kala (Time of administration), Kalavadhi (Duration) and Pathya-pathya ) of following drugs. | CAP | MK | KH |  | L&PP<br>T,DIS<br>,ML  | T-EMI,P-<br>VIVA           | F&S | III | V-RS,<br>H-SW |
| CO2 | Enlist active phyto-constituents & important formulations. Discuss research updates of following drugs.                                                                               | CK  | MK | K  |  | L&G<br>D,BS           | T-EMI,T-OBT<br>,P-VIVA,QZ  | F&S | III |               |
| CO8 | Enlist botanical name & family. Explain main Synonyms as per Bruhatryees and Bhavaprakasha. Vernacular name (Hindi, English and local name) of following drugs.                       | CK  | MK | K  |  | L&G<br>D,FC,<br>ML    | T-OBT,P-<br>VIVA,QZ        | F&S | III |               |
| CO8 | Demonstrate external morphology-habit, root, leaf, stem, flower, inflorescence, fruit, seed and officinal useful parts of following drugs.                                            | CC  | MK | KH |  | L_VC<br>,ML           | T-OBT,P-<br>VIVA,QZ        | F&S | III |               |
| CO7 | Describe varieties, grhyahrayatwa, adulterants, substitute and toxic effects of following drugs wherever applicable.                                                                  | CC  | MK | KH |  | L&PP<br>T,DIS<br>,EDU | T-EMI,T-OBT<br>,P-VIVA,PUZ | F&S | III | H-AT          |